

Technical Specification
For
Corporate Transaction Banking (CTB) Solution

Sl. No	Particulars/Features/Descriptions	Bidder Response	Remarks
1. System Architecture:			
1.1	System is built using a cloud-ready, modular, microservices-enabled, N-tier architecture, ensuring flexibility, scalability and ease of integration. Bidder should submit necessary documents, diagrams and illustrations having complete explanation of the system architecture including: a) End-user manual (Installation, operation and maintenance of server solution including database) b) Installation of solution on servers and clients c) Application Layer d) Configuration of the Solution e) Integration methods - e.g. API/Middleware, web services f) High Availability configuration and operation; failover tests g) Disaster Recovery Procedures h) Data flow diagrams with security and encryption descriptions i) Data dictionary j) Details description of all security parameters and configuration		
1.2	Hardware and environmental Platform (Please mention detailed specifications and quantities in a separate attachment document) a) Operating System (i.e. OS Independent, of UNIX/LINUX based will be preferable) b) Database servers (i.e. PostgreSQL /MySQL /Oracle are preferable) c) Specify all the Operating System, Database license requirements for Live operation. d) Specify the data replication method between DC and DR e) Specify the data backup strategy and frequency for the data backup considering lowest data loss. f) Compatible database software must support cluster in active-active and Active Passive mode, database replication server for disaster recovery, business continuity with seamless failover. g) Application (System must support both 32bit and 64-bit processors and operating system).		

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	h) Database Administration Tools- Specify which tools are provided by the vendor and which are provided by the database vendor. i) OS Administration Tools- Specify the tools provided by the contractor and which are provided by the system vendor. j) Specify the hardware (servers etc.) design that services have to be available in DC, DR and Far DR k) Please specify the network equipment, if any like hardware , Bandwidth , WAF , load balancer etc.		
2. System Support:			
2.1	Provide all general application support. Bidder must maintain a 24*7 help desk with adequate technical support personnel and infrastructure.		
2.2	System support shall support dynamic enquiries		
2.3	System support shall support remote monitoring and administration.		
2.4	System support shall support Internet Explorer, Mozilla Firefox, Google Chrome, Edge browser etc.		
2.5	System support shall support multi-language		
2.6	During UAT, along with test users, the system should have facility to enable tests to be run with simulated stored script in the Testing environment.		
2.7	Software licenses are installable, configurable, and operable for live operation, enabling service from any site such as DC, near DR, and far DR environments.		
2.8	The system should have a feature for setting up static codes or reference codes.		
2.9	System shall have facility to display Bank logo in the system at the time of running the solution and during viewing & printing the reports or statement		
2.10	The system shall have facility to provide administration tools and procedures for ongoing support and maintenance, including customization, of the proposed solution		
2.11	Protocol support like HTTPS, SSL		
2.12	System support for Local customization, Validation and screen design		

2.13	Secure network connectivity between Litigation and CBS/peripheral systems. Dedicated VLAN or secure segment recommended.		
2.14	Capacity of EOD configuration and data fetching or loading from CBS or other systems.		
2.15	Facility of real time interface and application programming interface.		
2.16	The system should have the facility to encrypt all the confidential information when being stored in the database and while being transferred through networks		
3. Integration:			
3.1	System shall have facility to allow Open-source architecture for integration of different systems in the bank including CBS		
3.2	Systems shall have capacity to integrate with another platform and shall support single sign in. It can be web and app-based banking open platform architecture to integrate existing and upcoming channels, in-house software and security solutions.		
3.3	Systems have facility to upgrade its field or logic or option for making the system compatible with change/upgrade bring in any interface for smooth operation.		
3.4	System shall have facility to integrate with following but not limited to: a) SMS server b) e-mail server		
3.5	System should have capacity to send reminders and notification through SMS and Email on set interval/on demand basis.		
3.6	System shall have an integrated facility for data warehousing Solutions.		
3.7	Integration methods Rest /SOAP APIs , Batch or ETL processing, SFTP and open API framework for local customization.		
4.Version Control			
4.1	Ability to restore previous versions in case of multiple versions.		
4.2	Notification of changes to relevant users.		
4.3	Existing data (if any) must be migrated to the new system.		
4.4	Data Gap / No Legacy data maps to proposed solution:		
5.Process Configuration:			
5.1	System shall have facility to support parallel processing of activity or flow where an application can be processed across more than one stage of activity		

6.Business Rules:			
6.1	The system shall have the capacity to incorporate a flexible, in-built business rules engine that allows bank administrators to: a. Configure and modify business rules or flow without vendor intervention including AI/ML Based False Positive Analysis b. Easily amend existing business rules or flow c. Implement a maker-checker approval process for rules or flow changes. d. Scenario Simulator user interface		
7.Restrictions for Generation and View			
7.1	System shall have facility to restrict any user/class/group (parameterized) for generation, authorization, view, report generation, etc.		
8. Deployment Requirements			
8.1	Bidder is requested to submit a complete deployment diagram with all interface requirements.		
8.2	An on-premises solution is preferred (Data Center & Disaster Recovery). Ideals for environments require full data sovereignty and control. Privet cloud /hybrid deployment is optional.		
8.3	Test, UAT, and Production environments must be configured.		
9.User Functionality			
9.1	License of the system shall be perpetual		
9.2	Comprehensive role-based access control (RBAC) with granular permissions and audit trails.		
9.3	All passwords are encrypted using industry-standard algorithms and securely stored in the database.		
9.4	System shall support to store password for each user in database in encrypted format		
9.5	System shall have facility to maintain password validation as per bank policy		
9.6	System shall have facility to support single sing on.		
9.7	The system shall have the facility to assign single or multiple roles to a user as per requirements.		
9.8	The system should have flexibility to change user role, approval authority time to time as per requirement.		
9.9	System shall have capacity to handle the access granting facility by administrator for user according to the assigned role &		

	privileges, user type –Maker and/or Authorizer (Admin User Define).		
9.10	Systems shall have the facility to access or restrict the user in different functions and options/functions are granted or revoked for each user as per assigned role.		
9.11	Integration with Active Directory or existing user management system.		
9.12	Audit trail and logs (view, edit, delete actions).		
9.13	Encryption of data at rest and in transit.		
9.14	The system shall have the facility to maintain user groups as per the bank's policy.		
9.15	System shall have capacity to set user Terminal IP (case to case basis), Operating Time, Application Server IP, etc.		
9.16	The system shall have facility for recovering passwords by the User himself/herself with bank approved process.		
9.18	The system shall have facility for automatically logging out/session timeout the users after a predefined period of being idle.		
9.19	System shall have facility to allow the administrators to manually lock/deactivate the users/ unlock/reactivate the users and all the events should be recorded.		
9.20	System shall have the facility to restrict the users to log into the system only from one workstation at a time		
9.21	The system should have a facility to allow the users to build queries and generate reports from database/ audit trails.		
9.22	System shall have facility for automatically locking next login if the maximum number (parameterized) of unsuccessful attempts is exceeded		
10. Disaster Recovery & Business Continuity:			
10.1	The system shall have facility for module-based redundancy and Business Continuity/Disaster Recovery plan including remote location.		
10.2	The system should be capable of maintaining the uptime of 100% using the disaster recovery site. Mention all documents describing the disaster recovery planning.		
10.3	System server and client software shall maintain its integrity in case of power failures and abrupt shutdowns.		
10.4	The system shall be capable of restarting and recovering after system failure with no loss of data or software components.		

10.5	System shall have the capability to identify the existence of program and/or system discrepancies and issue an alert to the system administrator.		
11. Security and Access Control:			
11.1	Role-based access control (RBAC).		
11.2	Integration with Active Directory or existing user management system.		
11.3	System should have facility to maintain hash values of the passwords when being stored in the database.		
11.4	System shall have capacity for being browser independent and must be compatible with latest version of browser, operating system and database		
11.5	Systems have the capacity to protect all kinds of security threats (Internal/External).		
11.6	Fully adheres to OWASP Top 10 standards, including protection against injection attacks, XSS, CSRF, and more, with regular penetration testing.		
11.7	Systems have a facility to generate reports indicating the success/failure of the batch processes. Any failure should be recorded with the failed item and the reason for failure.		
11.8	The system should have facilities for applications to handle all possible exceptional and erroneous events and appropriate user-friendly error messages should be displayed. i.e. server level messages should not be displayed to the end user		
11.9	System shall have facility to support secure communications utilizing SSL/TLS over HTTPS for secure data transfer.		
11.10	The system shall have facility for Users to query the logs for specific events and view filtered log reports, using a simple interface with fields for search and sorting parameters.		
11.11	The system should be compatible with any new technology or upgraded technology that is integrated with the system: a) Server b) Storage c) Operating system d) Database e) Network f) Security solution		
11.12	The Vendor must be capable of incorporate the relevant change immediately according to the recommendation of security advisories.		

11.13	System should have facility to identify and authenticate specific devices before establishing a connection		
11.14	The vendor should share the security recommendation of overall platform.		
11.15	System must be compatible to integrate with all type of ISSS (Industry Standard Security Solution)		
11.18	All software, libraries and components used by the system must have valid licenses and are up to date.		